

Q1

A) 3 ways to model subtypes:

- 1) Single table w/ a column indicating type

Vehicles

ID	Owner	Make	Type
0	Bob	Honda	Sedan
1	Alice	Toyota	Hatchback
2	Dave	Ford	Pickup Truck

- 2) Separate tables with a primary table with a shared primary key
Primary keys table

ID
0
1
2
3

Sedans

ID	Owner	Make
0	Bob	Honda
3	Deb	BMW

Hatchbacks

ID	Owner	Make
1	Alice	Toyota

Pickup_Trucks

ID	Owner	Make
2	Dave	Ford

3) Combination of the first 2: separate tables with common attributes in the primary table

Primary keys table

ID	Owner
0	Bob
1	Alice
2	Dave
3	Deb

Sedans

ID	Make
0	Honda
3	BMW

Hatchbacks

ID	Make
1	Toyota

Pickup_Trucks

ID	Make
2	Ford

B) analogy between entity subtype and subclasses in oop

The analogy between oop and entity subtypes is not entirely accurate because there are some differences between tables and classes. In OOP classes are essentially blueprints for creating classes and each object is an independent instance with its own state meaning that the requirement for deduplication and relational data separation isn't exactly required. However tables are different in that they don't have the same behavior as independent instances for each row but exist on the same table as a collection of rows to record data.

There are some similarities in that a subclass is conceptually inheriting data/attributes from its respective superclass and same with subtypes and supertypes in an EER diagram. The subtypes and subclasses then use specific attributes to distinguish them from each other.

To make the analogy more accurate we should introduce some of the functionality that classes have such as data transformation. Classes have functions/methods that can transform the data within that instance of the object so similarly tables should have functions that act upon the data within a table. Tables also need inheritance where the parent/super table's attributes are inherited by the child/sub table but to do so without duplication.

Q2)

A) Normalization benefits

- a) Normalization helps with reducing data redundancy which can improve consistency and reduce any possible insertion anomalies. Making a large table into smaller tables reduces any duplicated information amongst rows. A drawback of normalization is the increase in complexity of the database. By separating into multiple tables a single simple insertion becomes multiple insertions with multiple constraints to make sure the data is consistent. This results in a slower insertion query.

B) Show an incorrect table and how to fix a table

"Incorrect"/inefficient table:

Student ID	Student Name	Book Title	Book Author	Hours Read
1	John	The Adventures of Tom Sawyer	Mark Twain	10
2	Sarah	To Kill a Mockingbird	Harper Lee	5
3	Tom	The Adventures of Tom Sawyer	Mark Twain	8

A more efficient way to store the tables:

Student ID	Student Name
1	John
2	Sarah
3	Tom

Book ID	Book Title	Book Author
1	The Adventures of Tom Sawyer	Mark Twain
2	To Kill a Mockingbird	Harper Lee

Assignment ID	Student ID	Book ID	Hours Read
1	1	1	10
2	2	2	5
3	3	1	8

Q3

A) Example of data that is suitable for a single user DB?

- a) Data that is suitable for a single user DB would be a phone book or a login manager with username/email and password combo for a domain.

B) Why is structural dependence bad when storing data?

Illustrate structural dependence:

- a) Structural dependence is bad because it makes the DB schema more difficult to modify when the data changes. An example would be a table to store orders for a store. We have an initial table with all the information.

Order ID	Customer Name	Product Name	Quantity	Price
1	John Smith	Widget A	2	10.99
2	Sarah Jones	Widget B	1	7.99
3	John Smith	Widget A	1	10.99

But if we want to change the table so that we have a first and last name we are going to have to modify the data as well.

Order ID	First Name	Last Name	Product Name	Quantity	Price
1	John	Smith	Widget A	2	10.99
2	Sarah	Jones	Widget B	1	7.99
3	John	Smith	Widget A	1	10.99

There isn't much of a problem here but in the case that there exists another table that used the customer's name as a foreign key then this new change will break the connection/relation between the 2 tables because a single column is now 2 meaning that the foreign key needs to be a tuple requiring another change. Similarly other errors can occur across other tables.

C) Layered data abstraction

- a) Layered data abstraction provides several benefits. It helps to improve the overall structure and organization of the database by breaking down the large data into smaller layers thereby providing a layer of abstraction that makes it easier to modify and maintain databases over time

Q5

A) Pick 2 sites and describe ui actions

a) Instagram

i) UI action: search for profile

ii) Query:

(1) SELECT *

(2) FROM users

(3) WHERE username = 'zuckerberg';

b) Amazon

i) UI action: filtering search results by price range

ii) Query:

(1) SELECT *

(2) FROM products

(3) WHERE category = 'soldering iron'

(4) AND price BETWEEN 100 AND 500;

B) query to sql conversion

a) Typically the query to sql conversion occurs in the backend or api layer of the application stack. This way any security issues with the query can be detected and dealt with as well as keeping database access to trusted persons/entities so bad actors cannot attack your database.

Q6

- A) Most important phase in 2pl
 - a) The most important phase in 2PL for data access during transactions is the growing phase, during which locks are acquired on the required data items before any modifications are made to them. This phase is critical to ensure that the transaction can execute without conflicting with other transactions that may also need to access the same data items.
- B) When locks get released in 2pl what might cause the issue? Any fix?
 - a) If locks are released prematurely in 2PL, it can cause a phenomenon known as "lost updates". This occurs when two or more transactions attempt to modify the same data item concurrently, leading to one transaction overwriting the changes made by another transaction. To fix this issue, locks should be held until the transaction completes or aborts.
- C) What issue might arise if we do TM without locks? Any fix?
 - a) When performing transaction management (TM) without locks, there is a risk of conflicts between transactions that may access the same data items concurrently. This can result in inconsistencies in the data or incorrect results being returned. One way to fix this issue is to use a technique known as "optimistic concurrency control", which involves allowing transactions to proceed without locking the data, but checking for conflicts before committing the changes. If a conflict is detected, one of the transactions is rolled back and restarted.

Q7

A) Codd's relational operators

- a) Codd's relational operators lead to closure, which means that the result of a query on a relation is also a relation of the same type. This is advantageous because it allows us to chain multiple relational operators together to perform complex data processing tasks. For example, we can apply a selection operator to a relation to filter out unwanted data, and then apply a projection operator to the resulting relation to retrieve only the desired attributes.

B) Illustrate the four datatypes and list operations

- a) Vector:
 - i) Closure-preserving operation: Vector addition
 - ii) Non-closure-preserving operation: Scalar multiplication
- b) Matrix:
 - i) Closure-preserving operation: Matrix addition
 - ii) Non-closure-preserving operation: Matrix inversion
- c) Complex number:
 - i) Closure-preserving operation: Complex addition
 - ii) Non-closure-preserving operation: Complex division
- d) Color (with RGB components):
 - i) Closure-preserving operation: Color addition
 - ii) Non-closure-preserving operation: Color subtraction

Q9

A) Computed columns

- a) Computed columns are columns in a table whose values are automatically generated based on an expression or formula involving other columns in the same table. They can be used for various purposes such as generating summary statistics, deriving new columns, or validating data consistency.

B) Table of sines and cosines

- a) $\sin(\theta) + \cos(\theta) = 1$ (sum of $\sin(\theta)$ and $\cos(\theta)$ is equal to 1, then the formula is verified)

- i) Find the value of $\sin$ for θ degrees, i.e., $\sin(\theta)$.
- ii) Find the value of $\cos$ for θ degrees, i.e., $\cos(\theta)$.
- iii) Calculate the sum of $\sin(\theta)$ and $\cos(\theta)$.
- iv) Compare the result to 1.

b) Pearson correlation coefficient between columns X and Y

- i) Calculate the mean of X and Y, denoted by $\bar{X}$ and $\bar{Y}$, respectively.
- ii) Calculate the standard deviation of X and Y, denoted by S_x and S_y , respectively.
- iii) Calculate the covariance between X and Y, denoted by $\text{Cov}(X, Y)$.
- iv) Calculate the Pearson correlation coefficient, denoted by r , using the formula: $r = \text{Cov}(X, Y) / (S_x * S_y)$

Q8

- a. If the transaction coordinator fails during 2PC, we can use a backup coordinator to take over and complete the transaction. The backup coordinator should have all the necessary information to continue with the transaction.

- b. If a non-coordinator site fails partway through 2PC (between phase 1 and phase 2), we can use a timeout mechanism to detect the failure and abort the transaction if necessary. Each site should have a timer that starts when it sends its vote to the coordinator. If the site does not receive a decision from the coordinator before the timer expires, it assumes that the coordinator has failed and aborts the transaction.

- c. If a non-coordinator site fails at the start of 2PC (before phase 1), we can simply exclude it from the transaction. The coordinator can send a message to all sites to determine which ones are available to participate in the transaction. If a site does not respond, it is assumed to be unavailable and is not included in the transaction.